

Scenario: A university wants to predict if a student passes ($y = +1$) or fails ($y = -1$) based on hours studied (x_1) and hours slept (x_2).

- **Training Data:** One student.
 - Studied 2 hours, slept 1 hour: $x = [2,1]$.
 - **Actual Result:** Passed ($y = +1$).

Our Model: $f(x) = \text{sign}(w_1 \cdot x_1 + w_2 \cdot x_2)$.

Initial Weights: $w = [0.5, 0.5]$.

Learning Rate: $\eta = 0.1$.